

AGMU: Audio Generation Model Unlearning via Target Suppression and Retention

*Suppressing Forbidden Audio and Speech Concepts
While Preserving Retention*

Technical Report

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Abstract

We study *audio generation model unlearning*: given an audio model or audio processing pipeline and a target concept, suppress the tendency to produce or preserve that concept while maintaining behaviour on non-target inputs. We instantiate this in two regimes sharing a common evaluation principle: *report target suppression and retention together*.

- (1) **Acoustic-concept unlearning**, where a stochastic edit-policy modifies a real waveform so a downstream critic can no longer recognise the target concept. This track is trained with REINFORCE against a multi-term reward.
- (2) **Speech-word unlearning**, where SpeechT5 with LoRA adapters is evaluated for avoiding forbidden words using prompt rewrite, gradient ascent (GA), direct preference optimisation (DPO), and a preliminary ASR-reward RL prototype.

The speech benchmark is constructed around ten forbidden nouns, while the current headline run reports fully evaluated results for three words: *father*, *love*, and *world*. The results show that DPO provides the strongest current model-side forget–retain tradeoff: it reduces forbidden-word rate compared with the reference model while preserving retention substantially better than GA. GA suppresses aggressively only on one word in the current run and damages retention there; it does not generalise across the three evaluated words. Prompt rewrite remains the best inference-time oracle, while the preliminary RL prototype is a negative result rather than the headline method.

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1 Introduction

Modern generative audio models are trained on broad corpora that can contain content their operators do not wish to reproduce: copyrighted material, personal identifying information, harmful sound categories, or restricted spoken vocabulary. *Machine unlearning* studies how to make a trained model behave as if a designated subset of its training distribution had never existed, without retraining from scratch.

The image-generation community has developed a growing literature on concept erasure [Gandikota et al., 2023, Kumari et al., 2023]. The audio side is much younger. We argue that audio unlearning is interesting because it has two clearly distinct sub-problems:

- **Acoustic-concept unlearning** (“do not preserve rain”): the target is a sound category, evaluated by a waveform classifier.
- **Lexical unlearning in TTS** (“do not say *love*”): the target is a word, evaluated by ASR transcript matching.

The underlying tension is the same in both: *forget the target without destroying everything else*. This report introduces **AGMU** (Audio Generation Model Unlearning), a unified framework addressing both sub-problems. For clarity, the acoustic side is not yet unlearning a large pretrained text-to-audio generator. It is a stochastic waveform edit-policy trained to remove target acoustic evidence from real audio. The speech side is the main SpeechT5 benchmark for model-side word suppression.

Contributions.

1. A REINFORCE-based **edit-policy** for acoustic-concept suppression with explicit retention terms in the reward.
2. A **speech-word unlearning benchmark** built around ten high-frequency English nouns, with current headline results reported for *father*, *love*, and *world*.
3. A structured **baseline comparison** between the unmodified reference model, prompt rewrite as an inference-time oracle, GA, DPO, and a preliminary RL prototype for SpeechT5 word-level suppression.
4. A unified evaluation framing that reports forbidden-word suppression and retention together instead of treating target removal alone as success.

2 Related Work

2.1 Machine Unlearning

Machine unlearning in its modern form originates with Bourtole et al. [Bourtole et al., 2021], who proposed SISA training to make retraining cheap via data partitioning. Subsequent work distinguishes *exact* unlearning, where the unlearned model should match the distribution of a model retrained on retained data, from *approximate* unlearning, where cheap fine-tuning approximately removes forget-set influence [Sekhari et al., 2021]. Generative-model unlearning almost always falls in the approximate camp. Every result in this report therefore evaluates both axes: a *forget metric* and a *retain metric*.

2.2 Concept Erasure in Generative Models

In image generation, Gandikota et al. [2023] fine-tune cross-attention to suppress a concept while anchoring outputs on a retain set. Forget-Me-Not [Zhang et al., 2023] and Concept Ablating [Kumari et al., 2023] frame the problem as KL-regularised fine-tuning. UCE [Gandikota et al., 2024] and MACE [Lu et al., 2024] extend erasure to multiple concepts via closed-form weight edits. AGMU adapts the suppress-and-retain pattern from this literature to two distinct audio settings.

2.3 Preference Optimisation and Policy-Based Generative Methods

DPO [Rafailov et al., 2023] reformulates RLHF as a binary classification loss on preference pairs, removing the need for an explicit reward model. Diffusion-DPO [Wallace et al., 2024] and TANGO 2 [Majumder et al., 2024] apply preference optimisation to image and audio diffusion respectively. Their results suggest that DPO can be useful for audio-generation alignment at scale. Our setting is smaller and harder in a different way: the target is a single lexical item detected through ASR, and the preference signal is built from only a small number of generated TTS candidates. In our current speech benchmark, DPO is therefore best interpreted as a promising tradeoff method rather than as a complete lexical-erasure solution.

2.4 Audio and Speech Backbones

SpeechT5 [Ao et al., 2022] is the TTS backbone for our speech-word experiments. It takes a text prompt and a speaker xvector, then emits an 80-bin mel spectrogram vocoded by HiFi-GAN [Kong et al., 2020]. We adapt SpeechT5 using LoRA adapters [Hu et al., 2022]. **Whisper** [Radford et al., 2022] is used as the ASR evaluator. For acoustic-concept experiments we do not use a pre-trained text-to-audio model; the policy operates on real waveforms, isolating the unlearning signal from the large capacity of models like AudioLDM [Liu et al., 2023a], AudioLDM 2 [Liu et al., 2023b], or TANGO [Ghosal et al., 2023].

2.5 Datasets and Benchmark Construction

AGMU uses two dataset pipelines: a procedural acoustic-concept dataset and a LibriTTS-derived speech-word benchmark. The acoustic dataset contains three concepts: Rain, Wind, and Thunder, with 24 one-second clips per concept and 72 clips total. It is used to test whether a REINFORCE edit-policy can reduce target acoustic evidence under controlled conditions.

The speech-word dataset is the main benchmark for lexical unlearning. It is derived from LibriTTS `train-clean-100`. Only the transcripts are used directly as prompts; the evaluated audio is generated synthetically by SpeechT5 and then transcribed by Whisper. The benchmark target vocabulary contains ten high-frequency English nouns:

{love, house, water, fire, money, night, morning, father, mother, world}.

For each forbidden word, the split builder creates target and retain training and evaluation splits. Each word has up to 280 training prompts and up to 80 evaluation prompts. Target examples contain the forbidden word, while retain examples do not. This structure allows the benchmark to measure not only whether the model suppresses the forbidden word, but also whether it preserves non-forbidden speech content. The current headline run reports final averaged results for the three evaluated words *father*, *love*, and *world*; the remaining vocabulary is part of the benchmark design and future fuller evaluation.

DPO and GA do not train directly on the raw prompt rows. They use derived preference pairs built by rejection sampling. A frozen reference SpeechT5 model generates candidate

mels, Whisper transcribes them, and the shared reward function scores each candidate using forbidden-word absence and retention. The best candidate becomes the chosen sample, while the most-forbidden candidate becomes the rejected sample. Retain prompts are stored as anchors to stabilise training and prevent the model from damaging normal speech.

3 Problem Formulation

Let f_θ be a pre-trained audio generative model, or a real-audio edit pipeline in the acoustic case. Let C be a target concept, $\mathcal{D}_{\text{target}}$ inputs that elicit C , and $\mathcal{D}_{\text{retain}}$ inputs that do not. We seek $f_{\theta'}$ satisfying:

$$\begin{aligned} \mathbb{E}_{x \sim \mathcal{D}_{\text{target}}} [\text{ContainsConcept}(f_{\theta'}(x), C)] &\ll \mathbb{E}_{x \sim \mathcal{D}_{\text{target}}} [\text{ContainsConcept}(f_\theta(x), C)] && \text{(unlearn)} \\ \mathbb{E}_{x \sim \mathcal{D}_{\text{retain}}} [d(f_{\theta'}(x), f_\theta(x))] &\leq \epsilon && \text{(retain)} \end{aligned}$$

where `ContainsConcept` is a downstream detector, either an audio classifier or an ASR token match, and d is a perceptual distance. Trivial solutions such as silence or weight zeroing are excluded by the retention constraint. Methods are compared by their position on the forget–retain Pareto frontier. The reference model is used as the unmodified baseline, not as a strict lower bound on the forget metric.

4 System Architecture

The system decomposes into five layers shared across both tracks.

Data layer. The acoustic track uses procedurally generated Rain, Wind, and Thunder waveforms. The speech track uses LibriTTS utterances partitioned by forbidden word presence.

Critic/evaluator. The acoustic track uses a 3-layer Conv1D concept classifier with BatchNorm, GELU activations, and adaptive average pooling. The speech track uses Whisper-small as an ASR oracle.

Generator/policy. The acoustic track uses a stochastic convolutional `AudioEditPolicy` with FiLM conditioning. The speech track uses SpeechT5 with rank-16 LoRA adapters.

Training methods. REINFORCE with a multi-term reward is used for the acoustic edit-policy. The speech-word track evaluates prompt rewrite, GA, DPO, and a preliminary ASR-reward RL prototype.

Evaluation. Both tracks report `target suppression` and `retention` jointly. The speech track uses held-out target and retain splits. The acoustic track records target and non-target critic behaviour during in-loop evaluation.

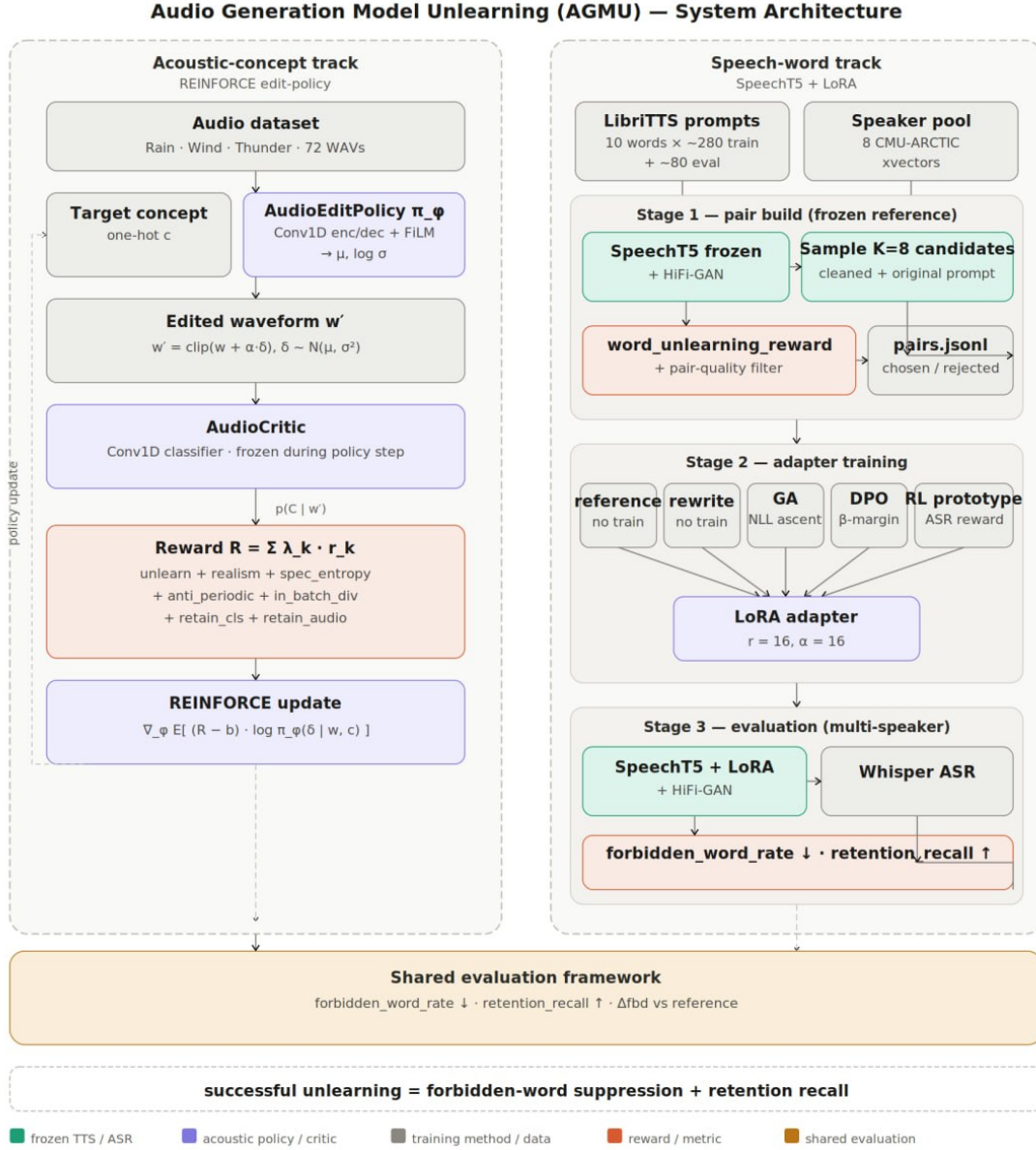


Figure 1: AGMU system architecture. Left: acoustic-concept track. A real waveform w and a target-concept one-hot vector c are passed to the **AudioEditPolicy** π_ϕ , which outputs a Gaussian residual policy through μ and $\log \sigma$. The edited waveform is $w' = \text{clip}(w + \delta)$, where $\delta \sim \mathcal{N}(\mu, \sigma^2 I)$. In the implementation reported here, the residual mean is bounded as $\mu = 0.6 \tanh(h_\mu)$; no additional post-sampling residual scale is assumed. The unlearning reward is computed from the **AudioCritic** score $1 - p(C | w')$ together with auxiliary realism, spectral, diversity, and retention terms. Right: speech-word track. The benchmark has three stages: pair construction, adapter training, and evaluation. During pair construction, a reference SpeechT5 model and Whisper build chosen/rejected mel pairs from rejection-sampled candidates. During adapter training, the system compares **reference**, **rewrite**, **ga**, **dpo**, and a preliminary ASR-reward **r1** prototype. During evaluation, SpeechT5 with the selected adapter generates speech with a multi-speaker pool, Whisper transcribes it, and the benchmark reports **forbidden_word_rate** and **retention_recall**.

5 Methods

5.1 Acoustic-Concept Edit-Policy

Let $w \in \mathbb{R}^T$ be a real waveform and c a one-hot concept vector. The policy π_ϕ outputs a Gaussian residual:

$$\mu = 0.6 \cdot \tanh(h_\mu), \quad (\mu, \log \sigma) = \pi_\phi(w, c), \quad \delta \sim \mathcal{N}(\mu, \sigma^2 I), \quad w' = \text{clip}(w + \delta, -1, +1).$$

The value 0.6 is the residual-scale bound applied to the mean residual, not a separate multiplier after sampling. This convention must remain consistent between the report and the implementation. The architecture is a convolutional encoder-decoder: a three-layer 1-D convolutional encoder, FiLM conditioning at the bottleneck, and a three-layer transposed-convolution decoder with separate μ and $\log \sigma$ heads.

Reward

The total reward is a weighted sum of seven components:

$$R = \sum_k \lambda_k \cdot r_k, \quad k \in \{\text{unlearn}, \text{realism}, \text{spec_ent}, \text{anti_per}, \text{div}, \text{retain_cls}, \text{retain_audio}\}.$$

- $r_{\text{unlearn}} = 1 - p_{\text{critic}}(C \mid w')$: the critic should not classify the edited waveform as the target concept.
- r_{realism} : RMS energy is kept in a plausible range.
- $r_{\text{spec_ent}}$: normalised spectral entropy penalises collapse to a single tone.
- $r_{\text{anti_per}}$: looped or repetitive content is penalised.
- r_{div} : minimum cosine distance to other batch elements penalises mode collapse.
- $r_{\text{retain_cls}}$: critic probability of the original label is preserved for non-target inputs.
- $r_{\text{retain_audio}}$: waveform-level edit magnitude is controlled for non-target inputs.

The retention terms are critical. Without them the policy can trivially suppress target evidence by damaging the whole signal, which is not meaningful unlearning.

Training

Policy and critic are trained jointly. The critic is supervised on (w, label) pairs after a warm-up stage. The policy uses REINFORCE with an EMA baseline:

$$\nabla_\phi J = \mathbb{E}[(R - b) \cdot \nabla_\phi \log \pi_\phi(\delta \mid w, c)].$$

An entropy bonus keeps the policy from collapsing to a deterministic edit.

5.2 Speech-Word Unlearning Methods

All speech methods share SpeechT5 + HiFi-GAN as the backbone and Whisper-small as the evaluator. GA, DPO, and RL use rank-16 LoRA adapters with $\alpha = 16$, dropout 0.05, applied to all SpeechT5 attention q , k , v , and out_proj matrices.

Baseline: Prompt Rewrite

The forbidden word is stripped from the input prompt before synthesis. This is *not* model unlearning because the model is unchanged, but it serves as a strong inference-time oracle. It achieves the strongest suppression because the target word is removed from the input.

Gradient Ascent (GA)

A LoRA adapter is trained to increase the model’s negative log mel-likelihood on a rejected forbidden mel while a retain-SFT anchor prevents drift:

$$\mathcal{L}_{\text{GA}} = -\lambda_{\text{ga}} \text{nlp}_{\pi}(p, m_{\text{rej}}, s) + \lambda_{\text{sft}} \text{nlp}_{\pi}(p_{\text{ret}}, m_{\text{ret}}, s_{\text{ret}}).$$

Gradient clipping is used to prevent collapse.

Rejection-Sampling DPO

Preference pairs are constructed by sampling K candidates per prompt and selecting a chosen candidate with the best reward and a rejected candidate with the strongest forbidden-word violation. The DPO objective on a regression mel decoder is:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma\left(\beta[(\text{nlp}_{\text{ref}}(c) - \text{nlp}_{\pi}(c)) - (\text{nlp}_{\text{ref}}(r) - \text{nlp}_{\pi}(r))]\right),$$

where c is the chosen mel and r is the rejected mel. We treat $-\text{nlp}$, the negative L1 mel loss, as a stand-in for $\log p_{\pi}(y | x)$. Reference NLPs are cached at pair-build time.

Preliminary ASR-Reward RL Prototype

The speech RL prototype samples audio from the current LoRA-adapted SpeechT5, transcribes it with Whisper, computes an ASR-level reward from forbidden-word absence and retention, and updates the adapter with a REINFORCE-like surrogate plus a retain anchor. This method is included for completeness because it is implemented and evaluated, but the current benchmark result is negative: it fails to reduce the forbidden-word rate. Therefore, it is not presented as the main method.

6 Experimental Setup

6.1 Acoustic-Concept Track

- **Concepts:** Rain, Wind, Thunder. The acoustic dataset contains 24 clips per concept, 72 clips total. Audio is 16 kHz mono and each clip is 1 s long. The clips are procedurally synthesised, which keeps the acoustic statistics controlled and makes the reward signal easier to interpret.
- **Critic:** 3-layer Conv1D + BatchNorm/GELU/AdaptiveAvgPool, trained with cross-entropy.
- **Policy:** convolutional encoder-decoder with FiLM conditioning and separate $\mu/\log \sigma$ residual heads.
- **Training:** critic warm-up followed by joint REINFORCE training. The code writes `metrics.csv` and `train_metrics.csv`; full quantitative acoustic results require running and collecting those files.

6.2 Speech-Word Track

- **Backbone:** microsoft/speecht5_tts + microsoft/speecht5_hifigan.
- **Forbidden-word vocabulary:** *love, house, water, fire, money, night, morning, father, mother, world.*
- **Current headline run:** final averaged results are reported for *father, love, and world.*
- **Source corpus:** LibriTTS train-clean-100; only transcripts are used directly as prompts.
- **Per-word splits:** up to 280 training prompts and up to 80 evaluation prompts per forbidden word, split into target and retain subsets.
- **Pipeline:** LibriTTS-derived splits, rejection-sampling DPO pair construction, LoRA training, and Whisper-based evaluation.
- **Exact benchmark run:** dpo_epochs=10, ga_epochs=6, rl_epochs=6, and eval_speakers=2.
- **LoRA:** rank-16 adapters with $\alpha = 16$ and dropout 0.05, applied to all SpeechT5 attention q , k , v , and out_proj matrices.

6.3 Training Hyperparameters

Table 1: Main hyperparameters used for the headline speech-word benchmark run.

Method	Hyperparameters
DPO	$\beta = 1.0$, num_epochs=10, learning rate $1e-4$, lora_r=16, sft_coef=0.4, retain_sft_coef=0.3.
GA	epochs=6, learning rate $1e-4$, ga_coef=2.0, ga_clip=1.5, retain_sft_coef=0.3, lora_r=16.
RL	epochs=6, learning rate $1e-4$, kl_coef=0.5, lora_r=16, num_target_prompts=80.

6.4 Metrics

We report two main metrics:

forbidden_word_rate (fbd) ↓

Fraction of Whisper-transcribed generated samples containing the forbidden word. Lower is better; 0 is perfect suppression.

retention_recall (ret) ↑

Fraction of non-forbidden reference tokens preserved in the Whisper transcript. Higher is better; 1 is perfect retention.

The difference from the unmodified reference model is used to interpret whether a method improves model-side suppression or makes it worse. The reference is the unmodified baseline. For forbidden-word rate, the main unlearning metric is the **target_eval** split. Averaging **target_eval** with **retain_eval** can be misleading because retain prompts usually do not contain the forbidden word by construction.

7 Results

7.1 Headline Target-Evaluation Results

The speech-word benchmark is evaluated using two main metrics: **forbidden_word_rate**, where lower values mean stronger suppression, and **retention_recall**, where higher values mean better preservation of non-forbidden content. These metrics must be interpreted together. A method is not successful if it lowers the forbidden-word rate only by damaging normal speech generation.

Table 2: Headline speech-word benchmark results on **target_eval**, averaged over the evaluated forbidden words: *father*, *love*, and *world*. Lower forbidden-word rate is better; higher retention recall is better.

Method	Target fbd ↓	Retention recall ↑	Interpretation
Reference	0.967	0.961	Unmodified baseline
Rewrite	0.000	0.917	Best prompt-side oracle, not model unlearning
DPO	0.839	0.922	Best model-side forget–retain tradeoff
GA	0.744	0.737	Lower mean fbd, but poor and unstable retention
RL	1.000	0.950	Negative prototype result

The table shows that prompt rewriting eliminates the forbidden word in the headline run, but this is achieved by modifying the input prompt rather than the model. Among actual model-side methods, DPO gives the best practical balance: its forbidden-word rate is lower than the reference model’s, and its retention recall remains close to the unmodified model. GA obtains a lower average target forbidden-word rate than DPO, but this average is misleading: it succeeds on one word while damaging retention and fails on the other two words. RL fails to reduce the target forbidden-word rate in this run.

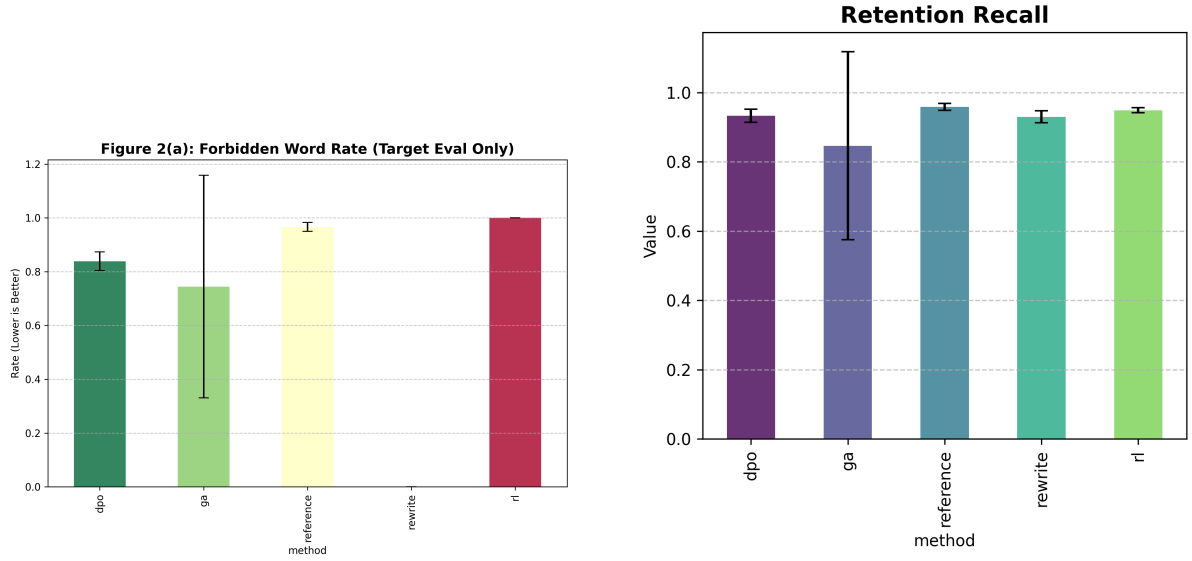
7.2 Per-Word Behaviour

Table 3: Per-word target forbidden-word rate. The GA result is bimodal: it suppresses *father* but does not suppress *love* or *world*.

Word	Reference	Rewrite	DPO	GA	RL
father	0.98	0.00	0.87	0.27 [†]	1.00
love	0.97	0.00	0.85	0.98	1.00
world	0.95	0.00	0.80	0.98	1.00

[†]For *father*, GA reduces forbidden-word rate but collapses retention substantially, with target retention around 0.29 in this run.

This table explains the wide error bars in the aggregate plots. GA is not uniformly mediocre; it is unstable. It performs strong suppression on *father*, but that suppression comes with severe retention damage. On *love* and *world*, GA remains close to the reference model on the forget metric. DPO does not produce perfect suppression, but its behaviour is more consistent across the three evaluated words.



(a) Target forbidden-word rate across methods. Lower is better.

(b) Retention recall across methods. Higher is better.

Figure 2: Main speech-word benchmark metrics. The forbidden-word-rate plot should use `target_eval` only, because this is where unlearning is measured. Rewrite achieves the strongest suppression because it removes the forbidden word from the input prompt, but it is not model unlearning. GA has a lower mean target fbd than DPO, but this is driven by one word and comes with severe retention loss. DPO keeps retention close to the reference model while still improving suppression, making it the best practical model-side tradeoff. Error bars show ± 1 standard deviation across the three evaluated words and reflect the per-word bimodality shown in Table 3.

7.3 Baseline Analysis

The baseline comparison reuses the same two metrics, `forbidden_word_rate` and `retention_recall`. The reference model establishes the starting point: it says the forbidden word almost every time on target prompts. Rewrite is the strongest suppression baseline, but it changes the input rather than the model, so it is not evidence of model unlearning. GA shows that directly pushing the model away from rejected forbidden examples can suppress a word, but the per-word table shows that this behaviour is unstable and can destroy retention. RL is included for completeness because it aligns training with the Whisper-based evaluation signal, but in the current run it is not competitive: its target forbidden-word rate remains 1.00.

DPO is therefore the strongest model-side method in the current run not because it has the lowest target forbidden-word rate, but because it gives the best forget–retain tradeoff. It improves over the reference model while keeping retention much closer to the original model than GA.

7.4 Acoustic-Concept Results

The acoustic-concept track is implemented as a controlled REINFORCE-based edit-policy experiment, but the final quantitative acoustic plots are not used as the main evidence in this report. The acoustic pipeline is designed to measure whether the edit-policy lowers the AudioCritic probability of the target concept after editing while preserving non-target audio behaviour.

Since the current report does not include final acoustic quantitative plots, we do not use this track to support the main empirical claim. Instead, the main quantitative evidence comes from the speech-word benchmark, where all methods are compared using `forbidden_word_rate` and `retention_recall`. Therefore, the acoustic-concept track should be interpreted as an implemented supporting pipeline rather than the main experimental result.

8 Discussion

Prompt rewriting is not unlearning. Rewrite achieves perfect suppression in the headline run because it changes the input, not the model. It is brittle to any input that bypasses the prompt filter, including typos, paraphrases, or code-switching. A model-side method robust under those conditions is the goal; rewrite is included as an inference-time oracle.

DPO is currently the strongest model-side tradeoff. DPO does not suppress as strongly as rewrite, and its average target forbidden-word rate is not lower than GA’s. However, GA’s lower mean is driven by one word and comes with a large retention collapse. DPO gives the best practical tradeoff among implemented model-side methods because it reduces forbidden-word rate compared with the reference model while preserving retention much better than GA. The improvement is real but modest, so DPO should be treated as a promising direction rather than a finished solution.

GA over-forgets and behaves unstably. GA can push the model away from forbidden outputs, but the same pressure can also damage normal speech generation. In the current headline run, GA suppresses *father* much more strongly than DPO, but that result is accompanied by very poor retention. For *love* and *world*, GA remains close to the reference model on target forbidden-word rate. This makes GA an unstable baseline rather than the best practical method.

The speech RL prototype is a negative result. The RL prototype is useful for research because it tests an evaluation-aligned training signal. In the current run, however, it fails on the forget metric and should not be described as outperforming DPO or closing the train/eval gap.

Mel-NLL is a weak proxy for ASR-level suppression. DPO and GA optimise mel-level proxy losses, while the benchmark measures ASR-level lexical behaviour. A small mel-spectrogram change near the formant transitions of the forbidden word can change Whisper’s transcript between variants such as *love*, *of*, or a noisy partial recognition, without producing a large change in mel L1 loss. DPO and GA may therefore see little useful gradient signal for changes that matter to the ASR evaluator. This mismatch helps explain why word-level TTS unlearning remains unstable.

Retention is non-negotiable. A method that destroys all speech synthesis capability can trivially reduce a forbidden-word rate. Our evaluation makes retention a hard constraint: methods must suppress the target while preserving useful non-target speech.

9 Limitations

1. **Dataset size.** The acoustic dataset contains 72 procedural clips, which is still small relative to typical audio benchmarks. The speech benchmark is also small relative to typical generative-model evaluations. Although the benchmark vocabulary contains ten forbidden nouns, the current headline quantitative run reports three words. Larger datasets such as ESC-50 and fuller LibriTTS splits are natural next steps.
2. **No human evaluation.** We rely on a downstream classifier and Whisper for the forget metric. A listening study would strengthen claims.
3. **One TTS backbone.** All speech experiments use SpeechT5. Results may not generalise to Bark, StyleTTS-2, or diffusion-based TTS systems.
4. **Single-concept unlearning.** Experiments unlearn one concept or one word at a time; multi-concept interference is not yet studied.
5. **Acoustic side edits real waveforms.** The edit-policy does not yet unlearn a pretrained text-to-audio generator in the strict sense. Lifting the reward onto LoRA adapters on AudioLDM, AudioLDM 2, or TANGO-style backbones is future work.
6. **Preliminary RL.** The speech RL prototype is not a successful method in the current run. It requires further work before it can be treated as a competitive SpeechT5 unlearning approach.
7. **DPO suppression is modest.** DPO improves over the reference model while preserving retention, but it does not eliminate the forbidden word. The current result should be read as a tradeoff improvement, not a complete lexical-erasure solution.
8. **Mel-level proxy mismatch.** DPO and GA optimise losses defined on mel spectrograms, while the target metric is based on Whisper transcripts. This mismatch can make lexical suppression noisy and unstable.

10 Conclusion

We presented **AGMU**, a unified framework for audio generation model unlearning instantiated in two regimes: acoustic-concept suppression with a REINFORCE-trained edit-policy and

speech-word suppression with SpeechT5 LoRA adapters. The central evaluation principle is to report unlearning and retention together. On the speech benchmark, rewrite is the strongest inference-time oracle, DPO is the strongest current model-side tradeoff, GA suppresses one word aggressively but behaves unstably and damages retention, and the preliminary RL prototype is a clear negative result. The DPO result is promising but modest: it improves the forget–retain balance rather than solving lexical unlearning completely. These findings show that word-level unlearning for TTS is not solved by directly transferring language-model preference methods. The main next steps are larger benchmark runs, stronger acoustic evaluation, human listening studies, and transferring the target-suppression objective to pretrained text-to-audio diffusion backbones.

Contribution Statement

All authors contributed to the project implementation, experimental analysis, and report preparation. The implementation work was divided across the acoustic-concept edit-policy pipeline, the SpeechT5 word-unlearning benchmark, baseline training, evaluation, and documentation.

References

- Ao, J., et al. (2022). SpeechT5: Unified-modal encoder-decoder pre-training. *ACL 2022*. <https://arxiv.org/abs/2110.07205>
- Bourtole, L., et al. (2021). Machine Unlearning. *IEEE Symposium on Security and Privacy*. <https://ieeexplore.ieee.org/document/9519428>
- Gandikota, R., et al. (2023). Erasing Concepts from Diffusion Models. *ICCV 2023*. <https://arxiv.org/abs/2303.07345>
- Gandikota, R., et al. (2024). Unified Concept Editing in Diffusion Models. *WACV 2024*. <https://arxiv.org/abs/2308.14761>
- Ghosal, D., et al. (2023). TANGO: Text-to-Audio Generation. <https://arxiv.org/abs/2304.13731>
- Hu, E.J., et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. *ICLR 2022*. <https://arxiv.org/abs/2106.09685>
- Kong, J., Kim, J., and Bae, J. (2020). HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis. *NeurIPS 2020*. <https://arxiv.org/abs/2010.05646>
- Kumari, N., et al. (2023). Ablating Concepts in Text-to-Image Diffusion Models. *ICCV 2023*. <https://arxiv.org/abs/2303.13516>
- Liu, H., et al. (2023). AudioLDM: Text-to-Audio Generation with Latent Diffusion Models. *ICML 2023*. <https://arxiv.org/abs/2301.12503>
- Liu, H., et al. (2023). AudioLDM 2: Learning Holistic Audio Generation with Self-supervised Pretraining. <https://arxiv.org/abs/2308.05734>
- Lu, S., et al. (2024). MACE: Mass Concept Erasure in Diffusion Models. *CVPR 2024*. <https://arxiv.org/abs/2403.06135>
- Majumder, N., et al. (2024). TANGO 2: Aligning Diffusion-Based Text-to-Audio Generations through Direct Preference Optimization. <https://arxiv.org/abs/2404.09956>

- Radford, A., et al. (2022). Robust Speech Recognition via Large-Scale Weak Supervision. *ICML 2023*. <https://arxiv.org/abs/2212.04356>
- Rafailov, R., et al. (2023). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. *NeurIPS 2023*. <https://arxiv.org/abs/2305.18290>
- Sekhari, A., et al. (2021). Remember What You Want to Forget: Algorithms for Machine Unlearning. *NeurIPS 2021*. <https://arxiv.org/abs/2103.03279>
- Wallace, B., et al. (2024). Diffusion Model Alignment Using Direct Preference Optimization. *CVPR 2024*. <https://arxiv.org/abs/2311.12908>
- Zhang, E., et al. (2023). Forget-Me-Not: Learning to Forget in Text-to-Image Diffusion Models. <https://arxiv.org/abs/2303.17591>